Karan Shihire

+91 8491832527 | karanshihire1@gmail.com | LinkedIn | GitHub | Pune, Maharashtra

Education

Modern College of Arts, Science and Commerce — Pune, Maharashtra
Bachelor of Computer Science (Minor: Embedded Systems & IoT) | CGPA: 8.0 — *Jul 2023 – May 2026*

Certifications

Microsoft Azure Machine Learning Fundamentals — Microsoft
Machine Learning Concepts, Azure ML Services, Responsible AI

Oracle Cloud Infrastructure Foundations Associate — Oracle
Cloud Architecture, OCI Core Services, Identity & Security

Projects

- **CareerLens — Semantic Job Recommendation Engine** — GitHub
  - Built an end-to-end AI system that parses PDF resumes, extracts technical skills, and generates semantic job recommendations using transformer embeddings.
  - Developed a hybrid recommendation engine combining cosine similarity and skill-overlap scoring to improve matching accuracy beyond embedding-only approaches.
  - Engineered a FAISS-based Top-K retrieval pipeline to efficiently search large job datasets and implemented a retrieval → reranking architecture for final recommendation generation.
  - Optimized recommendation latency by replacing exhaustive job comparisons with FAISS-based Top-K semantic retrieval.
  - Designed a modular FastAPI backend with reusable services for resume parsing, embedding generation, semantic retrieval, skill analysis, and recommendation scoring.
- **FaceTrack — Real-Time Attendance / Surveillance System** — GitHub
  - Built a real-time face recognition system capable of detecting and identifying multiple faces from live webcam streams using InsightFace and OpenCV.
  - Built face enrollment pipeline handling multi-angle capture and embedding extraction, reducing false match rate during testing
    * Challenge: matching faces reliably across lighting changes → solved using multi-angle enrollment + FAISS cosine similarity threshold tuning
  - Developed embedding similarity search workflow using FAISS to perform fast identity matching on high-dimensional facial embeddings.
  - Developed full-stack AI application integrating computer vision models, backend APIs, database systems, and responsive frontend interfaces.
- **NetWatch — Anomaly-Based Intrusion Detection System (IDS)** — GitHub
  - Developed an AI-driven intrusion detection system using Isolation Forest for anomaly detection on live network traffic data.
  - Built packet capture pipeline using Scapy that monitored TCP/UDP/ICMP traffic and flagged anomalies in under 100ms on local test environment
  - Built a real-time Flask dashboard for visualizing packet statistics, anomaly alerts, and live monitoring data.
  - Designed concurrent detection workflow for feature extraction, ML inference, and structured alert generation on real-time traffic streams.

Technical Skills

Languages: Python, SQL, Java

AI / ML: Computer Vision, Deep Learning, Scikit-learn, OpenCV, InsightFace, FAISS, NumPy, Pandas, NLP

Frameworks / Tools: FastAPI, PyTorch (Basics)

Platforms: GitHub, VS Code, Cursor, Render, Netlify , HuggingFace